

Teacher Role — Performance Audit & Optimization Report

Project: E-zentra (Zentra) **Scope:** Entire Teacher role — frontend, API, database, caching, realtime, skeletons, dialogs **Date:** October 6, 2026 **Mode:** Audit first, then implement. No rewrites, no redesigns — architecture, business logic, routes, schema, permissions, and styling conventions preserved.

Verification: `frontend tsc` clean · `frontend eslint` (teacher scope) 0 errors · `frontend build` clean · `backend typecheck` clean · `backend build` clean · `backend tests` 95/96 (the single failure is an `ocform01` Excel-template test that times out only under full-suite parallel load; it passes 8/8 in isolation and touches none of the changed code paths).

1. Teacher Pages Audited

All 23 routes under `/teacher/`, plus the shared shell and infrastructure:

URL	Source
<code>/teacher/overview</code>	<code>frontend/src/app/teacher/overview/page.tsx</code> (+ 12 <code>overview/components/*</code>)
<code>/teacher/overview/students</code>	<code>overview/students/page.tsx</code> + <code>student-list-data.ts</code>
<code>/teacher/overview/reports</code>	<code>overview/reports/page.tsx</code>
<code>/teacher/advisory/list</code>	<code>advisory/list/page.tsx</code>
<code>/teacher/advisory/attendance</code>	<code>advisory/attendance/page.tsx</code> + <code>AdvisoryAttendanceList.tsx</code>
<code>/teacher/advisory/schedule</code>	<code>advisory/schedule/page.tsx</code>
<code>/teacher/advisory/students</code>	<code>advisory/students/page.tsx</code> + <code>AdvisoryGradesTable.tsx</code>
<code>/teacher/advisory/students/[id]/academic</code>	<code>[id]/academic/page.tsx</code> + Academic components
<code>/teacher/advisory/students/[id]/attendance</code>	<code>[id]/attendance/page.tsx</code> + Attendance components
<code>/teacher/advisory/referrals</code>	<code>advisory/referrals/page.tsx</code> (server-paged + debounced) + 10 components
<code>/teacher/advisory/adm-cases</code>	<code>advisory/adm-cases/page.tsx</code> (server-paged) + components
<code>/teacher/classes</code>	<code>classes/page.tsx</code> + <code>MyTimetable.tsx</code> et al.
<code>/teacher/grading</code>	<code>grading/page.tsx</code> + <code>grading-data.ts</code>

URL	Source
/teacher/grading/[assignmentId]	[assignmentId]/page.tsx + ClassWorkspace , ScoreGrid , AssessmentList , dialogs
/teacher/attendance	attendance/page.tsx + AttendanceSheet , AttendanceRosterTable , attendance-taking-data.ts
/teacher/chat	chat/page.tsx (Suspense) + Bama* components
/teacher/anecdotal	anecdotal/page.tsx + Anecdotal components
/teacher/anecdotal/folders , /folders/student/[studentId]	folders pages
/teacher/grade-flags	grade-flags/page.tsx + FlagHistory , RaiseFlag* , ResolveFlagDialog
/teacher/schedule , /teacher/schedule/[sectionId]	schedule pages + WorkspaceCards , ScheduleWeekSetup , CatalogCards
/teacher/settings	settings/page.tsx + ProfileCard , PaletteCard

Shell: teacher/layout.tsx (TeacherShell : sidebar, notifications bell, adviser-claim gate, term badge, useTeacherRealtime() , queryClient.clear() on logout). **Stack:** TanStack Query (keepPreviousData / placeholderData , staleTime 15–60s) + axios apiClient (x-school-year-id / x-term-id headers, single-flight 401 refresh) + useSession + useTerm (termKey = schoolYearId:termId). Toasts via sileo / toast (components/ui/sonner.tsx , same backend). No loading.tsx / error.tsx under teacher/** — loading is inline skeletons + aria-busy shells. No router.refresh / location.reload anywhere in Teacher code.

2. Data Flow (verified per desk area)

Page → Component → Hook (termKey-scoped key, keepPreviousData) → apiClient (+term headers) → requireAuth → requireRole(subject_teacher, adviser) → teachable/ownership gate → batched prisma findMany → JS join / LRN-dedupe / sort → narrow-select payload → placeholderData cache → rendered UI (+ sileo toast on mutations)

Backend mounts serving Teacher: /api/teacher (3,258-line teacher.routes.ts : overview ? scope=critical|secondary|gradebook|full cached ttl:300 , student-list cached, schedule board, claim/verify/grant), /api/teacher/grading (workspace + components/preset/assessment CRUD), /api/teacher/grade-flags (paginated), /api/teacher/advisory (roster, enlist, advisee academic/attendance/anecdotal, marks, drawer, claim), plus /api/attendance (POST /bulk), /api/grades (score upsert, final lock), /api/anecdotal , /api/referrals/mine (cached), /api/notifications (take:50 , the realtime poll surface).

3. Critical Problems Found (all fixed)

ID	Severity	Problem	Fix
C1	Critical	<code>useTeacherInvalidate()</code> + realtime <code>TEACHER_KEYS</code> invalidated ~24–28 query prefixes on every mutation and every realtime event — one attendance save refetched schedule + grading + overview	Scoped invalidators; type-mapped realtime invalidation
C2	Critical	<code>POST /api/attendance/bulk</code> (subject + legacy paths): per-record <code>update/create</code> awaits, per-student <code>recomputeRisk</code> awaits, per-flagged-student <code>parentStudentLink.findMany</code> — all inside loops	Single <code>\$transaction</code> , parallel risk recompute, one batched parent-link query
H1	High	<code>attendance/page.tsx</code> rebuilt sorted/grouped <code>pairs</code> ($O(N^2)$) and <code>slotCards</code> sort/filter on every 60s clock tick and every search keystroke	<code>useMemo</code> + Map-based grouping + 250ms debounced search
H2	High	Split/stale query keys: <code>grade-flags/options</code> under two keys; <code>teacher-schedule-subjects/teachers</code> and <code>attendance-section-matrix</code> missing <code>termKey</code>	Unified keys, all term-scoped
H4	High	Assessment max-rescale: N per-score updates	Single <code>\$transaction</code> (refresh was already keyed + parallel)
M1	Medium	<code>FlagHistory</code> fired 3×100-row 7-join fetches on mount regardless of tab	Tab-gated queries; advisership via small shared sections query
M2	Medium	Un-debounced search in sheet/score/catalog/list filters	<code>useMemo</code> + shared <code>useDebounceValue</code>
M3	Medium	Sequential claim-release loop	<code>Promise.allSettled</code>
M4	Medium	Generic skeletons (anecdotal, advisory list, student list, roster table, week setup)	Geometry-matched skeletons

Deliberately untouched (already correct): no full-page reloads; `classDetailKey` teacher+term isolation; `keepPreviousData` back-nav; `markSelfNotified` echo suppression; `requestIdleCallback` prefetch; double-submit guards; `ClassWorkspaceSkeleton` + `attendance-skeleton` (used as conversion templates); weight/preset full-subject recompute (semantically required — weights affect everyone).

4. API Improvements

- `POST /api/attendance/bulk` (both subject and legacy AM/PM paths): N sequential write round-trips → **one atomic** `$transaction`; N sequential risk recomputes → `Promise.all`; N parent-link lookups → **one batched** `findMany ... WHERE studentId IN [...]`; records capped at 300 via zod (class-size bound + abuse cap).

- `GET /api/teacher/grade-flags` : fetch-all-then-slice-in-memory → **real skip/take + count** (run in parallel), with case-insensitive DB-side `q` across student name, LRN, and subject. Response shape unchanged (`{data, rows, total, unfilteredTotal, summary, page, totalPages, limit, pageSize}`); legacy bare-array shape preserved when no pagination params are sent.
- `PATCH /teacher/grading/assessments/:id` (max rescale): N per-score updates → **one \$transaction**.
- `GET /api/referrals/mine` **intentionally unchanged**: its sort key (`referredAt`) derives from the audit timeline rather than a column, and `?highlight=` deep-linking requires full ordering; per-teacher queues are small and the endpoint is already server-searched. Changing it would trade correctness for appearance.

5. Database Improvements

- Write batching above: one round-trip per attendance sheet instead of N (atomic as a bonus).
- **No new indexes added — verified unnecessary.** The schema already covers every hot path:
`AttendanceRecord` (`sectionId,subjectId,termId`), (`studentId,termId`), uniques on (`studentId|rosterId, subjectId, date, slot`); `StudentGrade` uniques on (`assessmentId, studentId|rosterId`); `FinalGrade` uniques + (`studentId,termId`); `GradeFlag` (`status,ownerId`), (`raisedBy`), (`studentId`).

6. Caching Improvements

- **Scoped invalidation** (`frontend/src/app/teacher/components/use-teacher-invalidate.ts`): `marks / grading / schedule / overview / referrals / advisory / anecdotal / flags / settings` (+ `all` reserved for logout-type resets). All 16 mutation call sites migrated to their scope; every scope still bumps `["teacher-notifications"]` so the bell badge stays live.
- **Scoped realtime** (`frontend/src/lib/realtime/teacherChannel.ts`): `Notification.sourceTable/type` → minimal affected prefixes (attendance → marks keys; score/final-grade → grading keys; timetable/teacher → schedule keys; referral → referral keys; anecdotal → anecdotal keys; roster/intervention → advisory keys; grade-flag → flags). Unknown events invalidate nothing but the bell badge. The 2s throttle, `seenIds` dedupe, `markSelfNotified` suppression, 5s poll transport (required — the anon Supabase client cannot receive row-scoped events), `document.hidden` skip, and focus repoll are unchanged.
- **Key normalization**: `["grade-flags","options",termKey]` unified; `termKey` added to `teacher-schedule-subjects`, `teacher-schedule-teachers`, and both `attendance-section-matrix` consumers. Prefix invalidation still matches; one-time cache bust is expected and safe.
- Term/user/role isolation preserved on every key (`teacherId, termKey`); `queryClient.clear()` on logout untouched.

7. Mutation Improvements

- Academic-record policy applied throughout: **per-row/cell spinner + Saving... button state, no optimistic flip** — `sileo.success` only after server confirmation, prior state + `sileo.error` on failure.
- Missing `if (saving)` return guards added to `ScoreGrid.handleSaveAll`, `WeightsDialog.handleSave`, `AddAssessmentDialog.handleSave` (previously `disabled` -only, double-Enter could double-fire).
- Every Teacher dialog now locks via `CardModal` `dismissable={!pending}` (Escape / overlay / X all disabled mid-flight).

8. Skeleton Improvements (position · width · height · spacing · structure verified per page)

Page	Before	After
Anecdotal repo	Generic centered 3-line pulse	Wired the existing-but-unused <code>skelPanel/skelFolderGrid/skelCard</code> CSS: same grid (<code>1fr + 17rem</code> rail), panel header + search + 6 folder cards, rail summary cards, responsive collapse
Advisory list	5× generic <code>h-10</code> rows	Same card + header/filter widths, 4 real columns (Student 220 two-line / Risk 130 / Factors 180 / Section 140), 8 rows, pager footer
Student list	2× generic blocks	Same heading + rail grid: table card (Student / Attendance / Academic Grade, badge-pair rows), section-rail cards
Roster table	5× generic <code>h-10</code> rows	Same card + filter/Blocks/Save header, 4 columns (Student 220 / Attendance 100 / Status 180 boxes / Meetups 360 dot strip), 8 rows
Week setup	Generic <code>h-8 + h-72 + pills</code>	Same bordered table: time gutter + 5 day columns, 8 <code>min-h-14</code> rows
Gradebook workspace, attendance sheet/rail, flag-history board, overview	Already geometry-matched	Untouched (used as templates)

Cached-data rule holds everywhere: skeleton only when `isPending` with no data; cached data renders instantly with background revalidation (existing `placeholderData / keepPreviousData` pattern extended, never regressed).

9. Dialog → CardModal Conversion (28 files, ~35 instances)

All `Dialog / AlertDialog` usage under `teacher/**` converted to the shared `CardModal` (`components/ui/CardModal.tsx` — `null` when closed, body-scroll lock, Escape + overlay-click, `dismissable` flight lock, `watchKey` scroll-hint refresh), following the in-repo precedents (`AdviserClaimGate`, `BamaFlowDialogs`, coordinator confirm dialogs):

- **Confirms:** `SubmitConfirmDialog`, `AttendanceSheet` `edit-confirm`, `AttendanceRosterTable` `save`, `ScoreGrid` `empty/range` notices, `AssessmentList` `delete`, `ReferralDeleteDialog`, `BamaChat` `delete-chat`, `MyTimetable` `leave-term`, `schedule/page` `clear-all`.
- **Forms:** `AddAssessmentDialog`, `WeightsDialog`, `SlotEntryDialog`, `ScheduleConfigDialog`, `CatalogCards` ×4 (add-teacher, add-subject, subject list, teacher list), `ScheduleWeekSetup` ×3 (note, unlock, clear), advisory enlist, `ReferralCancelDialog`, `ReferralComposer` `choice`, `RaiseFlagDialog`, `ResolveFlagDialog`.
- **Readers:** `ReferralTrackDialog`, `ReferralStepDialog`, `FlagDetailDialog`, `AllAssessmentsDialog`, `ClassDetailDialog`, `AnecdotalComposer`, `AnecdotalDetail`.

Mapping: `DialogTitle/Description` → `title/description` props, `sm:max-w-sm` → `size="sm"`, `sm:max-w-md` / default → `size="md"`, tall readers → `size="lg"` / scroll body, footers → right-aligned action rows, `onOpenChange` -pending locks → `dismissable={!pending}`. Overlay-only `.dialog` width classes retired in 7 CSS modules. Zero `Dialog*` / `AlertDialog*` imports remain under `teacher/**`. Side benefit: closed modals now render `null` (no hidden dialog DOM × ~35 instances). No copy, validation, or behavior changes.

10. Realtime Improvements

- Per-event scoped invalidation (Section 6) replaces the blanket 28-prefix refetch; the 2s burst throttle, toast-per-row semantics, and echo suppression are unchanged.
- Poll transport kept at 5s by necessity (documented anon-client constraint) with hidden-tab skip and focus repoll.

11. Before / After (architectural deltas)

Surface	Before	After
Any Teacher mutation	~24–28 prefix refetches	1 scope (≤6 prefixes) + bell badge
Realtime event	Full-desk refetch (throttled 2s)	Affected scope only (≤7 prefixes)
Attendance submit (60 students)	~60 sequential writes + ~60 sequential risk recomputes + K parent queries	1 transaction + 1 parallel recompute + 1 parent query
Grade-flag board mount	3 × up-to-100-row 7-join fetches	Visible tab only (+ tiny shared sections probe)
Assessment max-rescale (60 scores)	N updates	1 transaction
Attendance rail	Sort + filter rebuilt per tick/keystroke	Memoized; search debounced 250ms
26 dialog files	Radix Dialog/AlertDialog trees	Single CardModal overlay; <code>null</code> when closed
5 skeleton surfaces	Generic pulses/rows	Geometry-matched layouts

No timing numbers are claimed — improvements are structural (request counts, query counts, payload bounds, render frequency).

12. Remaining Bottlenecks (honest)

1. First-visit overview/schedule-board payloads stay large until those endpoints get DB pagination + select trimming (mitigated by `scope=` + 300s Redis cache).
2. 90-dot roster blocks (~5k spans at 60×90) are unvirtualized — acceptable at current page sizes; `@tanstack/virtual` only if sections exceed ~200 rows (deliberately deferred, package not installed).
3. ≤5s realtime latency is inherent to the poll transport; focus-repoll covers the tab-away case.

13. Validation

- `frontend: npx tsc --noEmit clean · npx eslint src/app/teacher src/lib/realtime/teacherChannel.ts`
0 errors · `npm run build clean` (all 23 teacher routes emit).
- `backend: npm run typecheck clean · npm run build clean · npm test 95/96` (single `ocform01` timeout under parallel load only; 8/8 in isolation; unrelated paths).
- Manual checks recommended on review: Overview → Attendance → Grading → Advisory → Student → Back navigation; term switch; logout/login cache isolation; dialog open/close/Escape/pending-lock on each converted modal.

14. Files Changed (session)

Frontend: `use-teacher-invalidate.ts` (scoped API) · `teacherChannel.ts` (scoped realtime) · `attendance/page.tsx` (memo + debounce) · `AttendanceSheet.tsx` (memo + CardModal) · `AttendanceRosterTable.tsx` (scoped invalidate + CardModal + skeleton) · `SubmitConfirmDialog.tsx` · `ScoreGrid.tsx` (guards + CardModals) · `WeightsDialog.tsx` · `AddAssessmentDialog.tsx` · `AssessmentList.tsx` · `AllAssessmentsDialog.tsx` (+CSS) · `ClassDetailDialog.tsx` (+CSS) · `MyTimetable.tsx` · `CatalogCards.tsx` (keys + CardModals + memo) · `CopiedSlotCard.tsx` (keys) · `ScheduleWeekSetup.tsx` (keys + CardModals + skeleton) · `SlotEntryDialog.tsx` · `ScheduleConfigDialog.tsx` · `schedule/page.tsx` · `advisory/list/page.tsx` (scoped invalidate + CardModal + skeleton) · `AdvisoryAttendanceList.tsx` (keys) · `teacher-overview-attendance-table.tsx` (keys) · `ReferralActionDialogs.tsx` · `ReferralComposer.tsx` (+CSS) · `ReferralStepDialog.tsx` (+CSS) · `ReferralTrackDialog.tsx` · `ReferStudentCard.tsx` · `use-reopen-referral.ts` · `referrals/page.tsx` · `BamaChat.tsx` · `AnecdotalChat.tsx` (keys) · `AnecdotalDetail.tsx` (+CSS) · `AnecdotalComposer.tsx` (+CSS) · `anecdotal/page.tsx` (skeleton) · `FlagHistory.tsx` (gating + probe + scoped invalidate) · `FlagDetailDialog.tsx` (+CSS) · `RaiseFlagChat.tsx` (keys) · `RaiseFlagDialog.tsx` (+CSS) · `ResolveFlagDialog.tsx` (+CSS) · `overview/students/page.tsx` (skeleton) · `settings/page.tsx` (parallel releases) · `ProfileCard.tsx` · `PaletteCard.tsx` . **Backend:** `attendance.routes.ts` (batched bulk ×2 paths, records cap) · `grade-flags.routes.ts` (DB pagination) · `grading.routes.ts` (batched rescale).

End of report.